

Trend Analysis Results Explanation for ResearchLens

This document explains the trend summary output shown in the dashboard, including:

- Trend Analysis Results
- Rising Topics
- Stable Topics
- Fastest Growing

1. What this summary means

The Trend Analysis Results section summarizes what Module 4 produced for the current dataset. It shows how many detected topics are classified as rising or stable, and identifies the topic with the strongest growth.

Example output

- 1 Rising Topics
- 4 Stable Topics
- segmentation / annotated / challenge
- Fastest Growing

The line `segmentation / annotated / challenge` represents a topic label or topic keywords that describe the strongest-growing topic.

2. How the counts are computed

The frontend calculates these summary values from `backendResult.modules.module4.trends`. Each trend object has a `trend` label:

- `rising`
- `stable`
- `declining`

The summary counts are:

```
Rising Topics = number of trends where trend === 'rising'  
Stable Topics = number of trends where trend === 'stable'
```

So **1 Rising Topics** means exactly one topic was classified as rising. **4 Stable Topics** means exactly four topics were classified as stable.

3. How "Fastest Growing" is chosen

The dashboard selects the fastest-growing topic by comparing a simplified growth rate across all topics.

For each topic, it computes:

```
growthRate = (maxCount - minCount) / maxCount
```

Where:

- `maxCount` is the highest yearly paper count for that topic
- `minCount` is the lowest yearly paper count for that topic

Then it picks the topic with the highest `growthRate` value. That topic is shown as the **Fastest Growing** topic.

4. How trend labels are assigned

Trend labels come from Module 4 in the backend. The backend computes a linear regression slope from yearly paper counts for each topic.

The thresholds are:

- `slope > 0.08` → **rising**
- `slope < -0.08` → **declining**
- otherwise → **stable**

So a topic is rising when its year-over-year publication count has a sufficiently positive slope. Stable topics are those with a flatter publication trajectory.

5. What the topic label means

The topic label displayed under **Fastest Growing** is derived from the trend topic name. In the example, `segmentation / annotated / challenge` is the topic name or summary label for the topic that had the highest growth rate.

6. Why this matters

- **Rising Topics** highlights current areas gaining momentum.
- **Stable Topics** shows areas with consistent publication output.
- **Fastest Growing** points to the topic with the strongest recent evidence of growth.

This summary helps users quickly identify which research directions are emerging and which are steady.

File saved

This explanation is saved in `trend-analysis-results-explanation.md` in the project root.